Course Outline

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TOPIC 1 : PROPERTIES & ATOMIC STRUCTURE

Essential Chemistry Skills

Properties of Matter

Nanomaterials

Atomic Structure

TOPIC 2 : BONDING AND INTERMOLECULAR FORCES

Introduction to Bonding

Introduction

1. **Describe the valency of an atom as a measure of the atom's bonding capacity, and use the periodic table to establish the valency of an atom for periods 1 to 3.**
2. **Explain how the ability of atoms to form chemical bonds is related to its electronic structure, particularly the stability of the valence shell.**

   An atom with very little valence electrons will tend to lose those valence electrons to obtain a full valence shell, forming a cation, while those with many valence electrons will tend to gain electrons to obtain a full shell, forming anions
3. **Explain how electrostatic attraction between oppositely charges species / regions leads to chemical bonds**

   Two Negative or positive charges repel each other, while negative charges attract positive charges. Therefore if there is a region of overall negative charge and one of overall positive charge, they will be attracted to each other. E.g. In an ionic bond, a negative ion is attracted to a positive ion
4. **Use the periodic table to establish the type of bonding present in elements and compounds.**

   Non-metals: covalent
   Metals: metallic
   Non-metal & Metal: ionic
5. **Explain why the type of bonding present in a substance defines the physical properties of that substance, including melting and boiling points, conductivity of heat and electricity, and hardness.**

   Different bonds have different properties and strengths due to the nature of their bonding. Ionic bonded substances are hard and brittle (like ions will repel each other if they are shifted even slightly) with high melting points (strong ionic forces). Covalent molecular substances have low melting points (only weak intermolecular forces need to be broken) and are non-conductors (no mobile charged particles). Metallic substances are malleable and ductile (held together by a sea of delocalised electrons which is non-directional) high melting point (strong electrostatic forces) and a good conductor (mobile electrons).

Ionic Bonding

1. **Define ions as atoms or groups of atoms that are electrically charged due to loss or gain of electrons.**

   Ions are atoms or groups of atoms positively or negatively charged due to loss or gain of electrons
2. **Describe and explain how ions are held together to form an ionic lattice.**

   Ions are held together in a 3D lattice structure, where each ion has 6 oppositely charged ions on each side of it. The opposite charges attract each other, and overcome the force of repulsion between like ions due to their closer distance.
3. **Use understanding of ionic bonding to explain the physical properties of ionic compounds, including:**

a. **High melting and boiling points**

Ionic bonds are very strong, and lots of heat energy is required to break them

b. **Hardness and brittleness**

Ions are held in specific positions in a lattice, so shifting part of the lattice will result in like ions being next to each other, and the repulsive forces between them will make the substance break

c. **Conductivity in solid, liquid and aqueous states.**

Solid: Non-conductor; No mobile charged particles as all the particles are held in specific positions in their lattice
Liquid/Aqueous: Conductor; The ions are not held in fixed positions and there are mobile charged particles able to move around

4. **Calculate percentage composition of an ionic compound from the relative atomic masses of the constituent species.**
5. **Define and describe the difference between anhydrous and hydrated salts, and explain the role of water of crystallisation.**

   A hydrated salt is a salt which contains water molecules, and an anhydrous salt is one which does not contain water molecules (can be made by heating a hydrated salt). Water of crystallisation is the water needed for a salt to maintain its crystalline properties
6. **Calculate the formula of an ionic compound from empirical data, including the formula of a hydrated salt.**
7. **Perform stoichiometric calculations in order to establish percentage composition and empirical formula of ionic compounds**

Metallic Bonding

1. **Describe how metallic bonding holds metal atoms together, in a lattice of positively charged ions surrounded by delocalised electrons**

   Metallic bonded substances are held in a network structure with metal cations held together by electrostatic forces of attraction between them and the sea of delocalised electrons - the metal atoms lost electrons to form stable valence shells, and the electrons are mobile and free to move through the metal (sea of delocalised electrons)
2. **Explain the physical properties of metals, including:**

   a. **malleability and ductility**

   The bonds are non-directional, meaning that the cations can move around and the bonds won't be broken. This allows metal to be beaten into sheets or pulled out into thin wires

   b. **thermal and electrical conductivity**

   Mobile electrons can move away from a positive terminal and towards a negative terminal to conduct electricity.
   When heated, the electrons start vibrating (kinetic energy) and they knock into other electrons as they move, creating a chain reaction that keeps going along the entire metal allowing it to conduct heat. The cations can also vibrate to contribute to its conductive nature.

   c. **generally high melting and boiling points**

   The electrostatic forces are very strong, lots of heat energy is required to break them

Covalent Bonding

1. **Describe a covalent bond as a shared pair of electrons, resulting in electrostatic attraction between positive nuclei of the atoms and the shared electron pair.**

   A covalent bond is the strong electrostatic force of attraction between a shared pair of electrons and the nuclei of atoms on either side
2. **Draw Lewis structures for covalent molecules, showing both bonding pairs and lone pairs of electrons.**
3. **Use the valence shell electron pair repulsion (VSEPR) theory along with Lewis structures to explain, predict and draw the shapes of molecules and polyatomic ions.**

   Tetrahedral: 109.5 degrees
   Pyramidal: 107 degrees
   Bent: 104.5 degrees
   Trigonal planar: 120 degrees

Bent: 119 degrees
Linear: 180 degrees

4. **Explain the physical properties of covalent molecular substances, including:**
    a. **low melting and boiling points**
        Molecules are only held together by weak intermolecular forces which don't require much heat energy to break
    b. **non-conductivity**
        There are no mobile charged particles as all electrons are localised and there are no ions
    c. **softness and brittleness**
        Weak intermolecular forces make the molecules easy to separate and break the substance (brittle) as well as easy to mould as the molecules are not arranged in a fixed lattice
5. **Recognise some common covalent network substances, including diamond, graphite and silicon dioxide.**
6. **Describe the range of allotropes of elemental carbon (diamond, graphite and fullerenes), and explain why they have significantly different physical and chemical properties.**
    Diamond: 4 covalent bonds per atom, very high melting/boiling point, non-conductor
    Graphite: 3 covalent bonds per atom, last electron is delocalised *within the layer* of graphite, conductor as there are mobile charged particles
    Fullerenes (e.g. Carbon nanotubes): Single layer of graphene rolled up into a cylinder; semi-conductor to super-conductor based on symmetry and shape, up to 100x as strong as steel and 0.25x density due to the continuous array of covalent bonds
7. **Explain the physical properties of covalent network substances, including:**
    a. **high melting and boiling points**
        Very strong covalent bonds require a lot of heat energy to break
    b. **hardness and brittleness**
        Very hard as the atoms are in fixed positions within the lattice and cannot move as the bonds are directional. Once an atom moves, its bond is broken and that part of the lattice will break (brittle)
    c. **thermal conductivity**
        Not a thermal conductor as there are no mobile particles
    d. **electrical conductivity**
        Not an electrical conductor as there are mobile charged particles

Intermolecular Forces

1. **Describe the difference between intramolecular bonding and intermolecular forces**
    Intramolecular bonding refers to the bonds holding atoms within a molecule together (covalent bonds). Intermolecular bonding refers to the weak electrostatic forces between molecules, holding covalent molecular substances together.
2. **Compare the relative strength of ionic, covalent and metallic bonding with intermolecular forces**
    Intermolecular forces are 10-100 times weaker than covalent bonds
    Strongest to weakest: Covalent, Ionic, Metallic; Ion-Dipole, Hydrogen, Dipole-Dipole, Dispersion
3. **Recognise that intermolecular forces are present in all substances, but are most significant in discussion of the properties of covalent molecular substances, due to the lack of any stronger interactions between molecules**
    All substances have intermolecular forces to some degree (e.g. Dispersion forces due to random movement of electrons), but as they are so weak compared to other types of bonding they are insignificant when discussing properties of non-covalent-molecular substances
4. **Describe how dispersion forces arise in all substances, and explain how molecular shape, and number of atoms affect the strength of dispersion force**
    Dispersion forces are the temporary, fluctuating dipoles created due to the random movement of electrons. They occur in all substances as electrons always have some degree of random movement, which creates temporary dipoles in a particle and induced dipoles in neighbouring particles.
    Larger molecule size means there are more electrons in the molecule, which can produce a stronger dispersion force as they can create a larger dipole
    A molecule shape with more surface area (e.g. Carbon chain vs tetrahedral) will be able to create stronger dispersion forces as there is more area for it to attract to neighbouring molecules

5. **Use knowledge of electronegativity to establish bond polarity**
   The more electronegative atom is slightly negative, other slightly positive. It is only considered a dipole/polar bond if the electronegativity difference is greater than 0.4
6. **Use understanding of molecular shape, bond polarity and symmetry to establish and explain the polarity of molecules**
   With dipole bonds:
      Symmetrical molecules are non-polar overall, as opposing dipoles 'cancel each other out'
      Non-symmetrical molecules will be polar, slightly positive/negative ends correspond to the bonds
   If there are dipole bonds of different strength the greater dipole will 'override' the smaller dipole
   No dipole bonds = non-polar
7. **Describe how dipole-dipole forces arise, and explain how the relative size of a molecular dipole affects the strength of dipole-dipole forces**
   Dipole-dipole forces occur when the slightly negative end of a polar molecule is attracted to the slightly positive end of another polar molecule, and vice versa. A relative molecular dipole makes the dipole-dipole bond stronger, as the molecule has greater slightly negative/positive ends which can form stronger attractions.
8. **Describe how hydrogen bonding arises, end explain why this special case of dipole-dipole bonding is so much stronger than other examples of dipole-dipole bonding**
   Hydrogen bonding is the strong dipole-dipole attraction between an electron deficient hydrogen atom bonded to a highly electronegative N, O or F atom, and a lone pair of electrons on another highly electronegative N, O or F atom bonded to another hydrogen atom on a different molecule.
   It is so much stronger than other forms of dipole-dipole bonding as the dipoles created by the H with N/O/F are relatively very large, meaning that the H is more positive than usual in a dipole, and it is attracted to a lone pair of electrons rather than just a slightly negative dipole.

   Why don't hydrogen bonds occur in non-polar molecules?
9. **Draw diagrams to show how all types of intermolecular forces occur between neighbouring molecules**
10. **Compare the relative strength of the types of intermolecular forces**
   Strongest to weakest: Hydrogen, Dipole-Dipole, Dispersion
11. **Use understanding of bond polarity, molecular shape and symmetry to establish the nature and strength of intermolecular forces present in a covalent molecular substance**
   Use bond polarity and molecular shape to determine whether the molecule is polar or not. If it is non-polar, it will only have dispersion forces. If it is polar, it will have dipole-dipole forces. If it has H-N, H-O or H-F arrangement of atoms, it will have hydrogen bonds.
12. **Explain how the physical properties of covalent molecular substances depend on the strength of intermolecular forces, including:**
   a. **melting and boiling point**
      Stronger intermolecular forces = more heat energy required to break them and melt/boil the substance (melting/boiling only involves breaking intermolecular forces, not intramolecular forces)
   b. **vapour pressure**
      Stronger intermolecular forces = Lower vapour pressure at the same temperature as a molecule with weaker IMF. Stronger IMF means that the molecules will be attracted more strongly to each other, decreasing their kinetic motion and reducing vapour pressure.
   c. **solubility in water and organic solvents**
      "Like dissolves like" - Strength of solute-solvent attraction must be enough to overcome solute-solute and solvent/solvent attraction
      Therefore: Solutes which form dipole-dipole forces and hydrogen bonds generally dissolve well in water; Non-polar solutes generally dissolve well in organic solvents such as benzene (C_6H_6)

      Polar solutes will generally dissolve in polar solvents, only form dipole-dipole attraction which are relatively the same strength
      Non-polar solutes generally dissolve in non-polar solvents, as they all only form dispersion forces which have relatively the same strength
      Non-polar solute won't dissolve in polar solvent as they can only form dispersion forces which aren't strong enough to overcome dipole-dipole forces in solvent
13. **Use understanding of intermolecular forces to explain patterns or relationships in melting and boiling points of covalent molecular substances**

Chromatography

1. **Describe the general principles of chromatography, including the role of the**

stationary and mobile phases

All chromatographic techniques use two phases: mobile and stationary phase. Often, one will be polar and one will be non-polar so that more polar substances will move either faster or slower through the stationary phase, separating the sample based on polarity

The mobile phase moves through/across the stationary phase, carrying the sample with it, separating it into constituent components as explained above

2. Explain the importance of intermolecular forces in the chromatography process
3. **Explain how chromatography can be used to separate and identify the components of a mixture**

   Below
4. Use knowledge of the polarity of the components of a mixture to be separated to select a chromatographic method for separation

   Don't they all separate based on polarity except GC which is based on BP
5. **Apply the general principles of chromatography apply to the following techniques:**
   a. **Paper chromatography (PC)**

      Separates based on polarity

      SP: Chromatography paper; MP: Liquid solvent

      The sample is spotted on a pencil line about 1.5-2cm above the bottom of the paper. The paper is placed in a container with the mobile phase up to about 1cm so that it doesn't touch the pencil line. The mobile phase moves up the chromatography paper through capillary action (adhesive and cohesive forces). When it has nearly reached the top, the paper is taken out of the container, and allowed to dry.
   b. **Thin-layer chromatography (TLC)**

      Separates based on polarity

      SP: Layer of silica gel/alumina on glass plate; MP: Liquid solvent

      Exactly the same as paper chromatography, except the plate is rigid so easier to control, and the separated constituent substances can be collected afterwards

      Substances from both PC and TLC can be identified using the Retention Factor (ratio of distance moved by substance vs distance moved by mobile phase)
   c. **Gas chromatography (GC)**

      Separates based on mass

      SP: Silica gel/Alumina on inside of long column; MP: Inert (unreactive) gas such as He or Ar

      Sample is vaporised and pushed through a long, narrow column coated with a polar substance by an inert gas (non-polar). Substances with higher boiling points will be closer to their liquid state than substances with lower boiling points, so won't be carried as easily by the carrier gas and will therefore move through the column slower. Additionally, more polar substances will move slower as they will be attracted more to the silica/alumina on the sides of the column, helping to separate substances with similar boiling point. The temperature of the tube is often gradually increased so that higher boiling point molecules will be released in a reasonable amount of time.
   d. **High-performance liquid chromatography (HPLC)**

      Separates based on polarity

      SP: Very fine solid particles (e.g. silica/alumina); MP: Liquid solvent

      If SP polar MP non-polar, and vice versa

      A column (often steel) is packed tightly with the stationary phase. A tiny amount of liquid sample is injected into the stream of mobile phase, which is being pushed through the column using a high pressure pump to speed up the process. In Normal Phase HPLC, the stationary phase is polar and mobile phase is non-polar, so less polar substances will dissolve better in the mobile phase and move faster through the column, achieving separation. Reverse Phase HPLC is the opposite, with the stationary phase being non-polar and the mobile phase being polar.

      Both HPLC and GC have detectors at the end which graph the results on a chromatogram, which shows the retention time of constituent substances of the mixture.

Solubility

1. **Explain the unique properties of water using understanding of its molecular shape and ability to form hydrogen bonds, including:**
   a. **Relatively high melting and boiling points**

      More heat energy is required to break the relatively strong hydrogen bonds in water than other Group 16 hydrides
   b. **Relative density of solid and liquid states**

In the liquid state, there are typically 1-2 hydrogen bonds per molecule. In the solid state, the molecules arrange themselves to have 4 hydrogen bonds per atom, leaving a more open structure.

c. **Surface tension, viscosity, cohesive and adhesive forces**

Surface Tension: The strong hydrogen bonds pull molecules on the surface towards the centre, so water is very resistant to increasing surface area

Viscosity: Strong hydrogen bonds don't break easily, so it doesn't flow very easily

2. **Classify solvents as polar or non-polar, and solutes as ionic, polar and non-polar**

Water (Polar Solvent)	**C_6H_{12} (Non-Polar Solvent)**
CH_3COCH_3 (polar section)	C_6H_6 (non-polar)
CH_3COOH (polar; only small hydrophobic section)	CH_3COCH_3 (non-polar section; dissolves better in water)
NH_3 (polar)	
HCl (polar)	
NaCl (Ionic)	

NOTE: Watch out for insoluble solvents

3. **Identify and describe interactions between solvent and solute, and use this to predict whether a solution will form**

Polar solvent + ionic solute: Ion-Dipole forces - stronger than hydrogen bonds
Polar solvent + polar solute: Dipole-dipole forces
Non-polar solvent: Dispersion forces
If the solvent-solute forces are strong enough to overcome the solute-solute and solvent-solvent forces the solution will form

4. **Describe and explain how ion-dipole forces form in solutions, and use this to explain why ionic solutes are often soluble in water or other polar solvents**

Polar water molecules form electrostatic forces of attraction with the ions. These are quite strong (stronger than hydrogen bonds) and if enough water molecules form these bonds with an ion, they are enough to overcome the ionic forces holding the ion to its compound, and it is dissolved in the solution. As these are strong attractive forces, ionic solutes are generally soluble in polar solvents.

5. **Describe solutions using the terms saturated, unsaturated or supersaturated, and use these ideas to describe the concentration of a solution**

Unsaturated: A solution in which more solute may be dissolved at a particular temperature

Saturated: No more solute may be dissolved in the solution at a particular temperature

Supersaturated: An unstable solution in which more solute is dissolved than is normally possible at a particular temperature. If disturbed, the solute will crystallise leaving a saturated solution

6. **Describe the concentration of a substance using moles per litre of solution (mol L^{-1}), mass per litre of solution (g L^{-1}) or parts per million (ppm)**

Moles per litre (mol L^{-1}) $c = \frac{n}{V}$

Grams per litre (g L^{-1}): $c = \frac{m}{V}$

Parts per million (ppm): $c = \frac{m\ (solute\ in\ mg)}{m\ (solvent\ in\ kg)} = \frac{m\ (solute\ in\ g)}{m\ (solvent\ in\ g)} * 1000\,000$

7. **Convert concentrations between different units (listed above)**
8. **Describe and explain the relationship between solubility and temperature, for both solids and gases**

An increase in temperature makes the particles of solute solvent have more kinetic energy:

Solubility of solids *generally* increases (higher kinetic energy makes the bonds between them easier to break)

Solubility of gases decreases (higher kinetic energy causes molecules to break intermolecular forces and escape the solution)

9. **Draw, interpret and analyse solubility graphs for solids and gases**
10. **Use understanding of solubility, along with the data tables, to predict whether solutions will form a precipitate**
11. **Write equations to show only the reacting species and products made in a reaction, including net ionic equations for precipitation reactions**

Net ionic equation = remove spectator ions
Always simplify coefficient ratio

12. **Give observations for reactions involving solutions, solids and precipitates, with colours of solutions and solids, using the Data Booklet**
    Solutions are always clear, but need to specify colour/colourless
13. **Conduct experimental work safely, competently and methodically to determine solubility of ionic compounds**
14. **State that potable drinking water must undergo various treatment processes to ensure it meets regulations for safe levels of solutes, including heavy metals**
    a. Aeration: Water sprayed into the air to reduce presence of dissolved gases such as CO_2
    b. Flocculation: Small suspended particles are made to clump together into larger particles which eventually sink to the bottom due to their weight. Achieved by adding alum ($Al_2(SO_4)_3$) and Calcium Hydroxide which precipitate $Al(OH)_3$, a gelatinous substance which suspended particles adsorb to
    c. Settling: Mixture left for some time to allow floc and gathered particles to sink to the bottom, then are removed
    d. Filtration: Water filtered through a bed of sand on gravel to remove any remaining suspended matter
    e. Chlorination/Clarification: Chlorine added to water to remove biological contaminants
    f. Fluoridation: Fluoride added (in the form of compounds) to increase fluorine concentration as it reacts with enamel to help prevent tooth decay
15. **Explain why drinking water must be monitored for safe levels on solutes**
    Many solutes are hazardous to human health, such as pesticides from agricultural runoff, heavy metals from rainfall runoff on roads (zinc in car tyres/brake linings), and pathogens from animal waste. All of these must be removed to a certain level to ensure that the water will be safe to drink (potable)

TOPIC 3 : ACIDS & STOICHIOMETRY

Acids and Bases

1. **Use the Arrhenius model to define and describe acids and bases in aqueous solution**
   Arrhenius Definitions:
   Acid: A species which increases H^+ ion concentration in aqueous solution
   Base: A species which increases OH^- ion concentration in aqueous solution
2. **Describe the common chemical and physical properties of acids and bases**
   Acids: Taste sour, sticky
   Bases: Taste bitter, slippery
3. **State that indicators can be used to identify acids and bases in solution, and use observations of indicator colour to classify solutions as acidic, basic or neutral**
   Universal indicator:

Acidic	Red
	Yellow
Neutral	Green
	Blue
Basic	Purple

4. **Write equations to show how acids form H^+ ions and bases form OH^- ions in solution**
   (Net ionic equations)
   Strong acids/bases ionise/dissociate completely, single arrow
   Weak acids/bases ionise partially, reversible arrow
5. **Distinguish between the processes of ionisation, where a covalent molecule splits into ions, and dissociation, where an ionic compound dissolves in water**
   Ionisation: Molecule splits forming ions
   Dissociation: Ionic compounds divide leaving individual ions
6. **Describe and explain the difference between strong and weak acids and bases, using discussion of the degree of ionisation or dissociation**
   Strong acids are fully ionised, strong bases are fully dissociated/ionised (more H^+/OH^- per mole)
   Weak acids are partially ionised, weak bases are partially ionised/dissociated (less H^+/OH^- per mole)

7. **Write equations to show the difference between strong and weak acids and bases**
   Single arrow for strong
   Double arrow for weak
   In ionic equations, DON'T SPLIT WEAK ACIDS/BASES AS THEY ARE MOSTLY PRESENT AS MOLECULES
8. **Identify some common strong acids, including HCl, HNO_3, and H_2SO_4, and some common weak acids, including CH_3COOH, H_3PO_4 and other organic acids**
   Strong acids: HCl, HBr, HI, HNO_3, H_2SO_4, $HClO_3$, $HClO_4$, HIO_4
   Weak acids: Everything else
9. **Identify some common strong bases, including group 1 and 2 oxides and hydroxides, and some common weak bases, including NH_3 and Na_2CO_3**
   Strong bases: Group 1&2 Hydroxides
   Weak bases: Everything else
10. **Differentiate between the terms strong and concentrated when used in reference to acids and bases**
    Strong/Weak refers to the degree of ionisation/dissociation
    Concentrated/Dilute refers to the concentration of the acid/base in solution (moles of solute per litre of solution)
11. **Describe and give observations for the reactions of acids and bases, including the reactions between:**
    a. **acids and metals**
       Acid + Reactive Metal → Salt + Hydrogen Gas
    b. **acids and metal oxides**
       Acid + Metal Oxide → Salt + Water
    c. **acids and metal hydroxides**
       Acid + Metal Hydroxide → Salt + Water
    d. **acids and metal carbonates and metal hydrogencarbonates**
       Acid + Metal Carbonate/Hydrogen-carbonate → Salt + Water + Carbon Dioxide
    e. **bases and ammonium salts**
       Base + Ammonium Salt → Ammonia Gas + Salt + Water
    f. **bases and non-metal oxides**
       Base + Non-Metal Oxide → Salt + Water
12. **Write word equations, molecular equations and ionic equations for the reactions listed above**
13. **Predict observations for the reactions listed above, clearly describing the reactants and products, any colours involved (see Data Booklet) and any changes observed**
    Remember to check if the metal is reactive in an Acid + Metal reaction
    Check colour exceptions
14. **Define the pH scale as a measure of hydrogen ion concentration, and state that it is an inverse logarithmic scale**
    $pH = -\log[H^+]$
15. **Use pH values to classify a solution as acidic, basic or neutral**
    pH < 7: Acidic
    pH = 7: Neutral
    pH > 7: Basic
16. **Describe the relationship between pH, acidity and alkalinity of a solution**
    Lower pH = higher acidity, lower alkalinity
    Alkalinity: A measure of the acid-neutralising capacity of water
17. **Define pH as $-\log_{10}[H^+_{(aq)}]$, and use this relationship to calculate pH from hydrogen ion concentration, and vice versa**
    $pH = -\log[H^+]$; $[H^+] = 10^{-pH}$
18. **Define the ionic product of water, K_w, as the product of the hydrogen ion and hydroxide ion concentrations, and state that $K_w = 1 \times 10^{-14}$ at 298 K**
    $K_w = [H^+][OH^-]$
    In water, $[H^+] = [OH^-] = 10^{-7} mol\ L^{-1}$, therefore $K_w = 10^{-14}$ at 298K
19. **State that a neutral solution is one where H^+ ion and OH^- ion concentration are equal**
    Neutral solutions have an equal H^+ and OH^- ion concentration
20. **State that the concentrations of H^+ and OH^- are both 1×10^{-7} mol L^{-1} at 25 °C**
21. **Use K_w and the pH expression to calculate the pH of a solution of a strong base**
    $[H^+] = K_w / [OH^-]$
    If $[OH^-]$ concentration is known, use above equation to find pH
    pH and pOH will equal to 14 if $K_w = 10^{-14}$, 20 if $K_w = 10^{-20}$, etc.
22. **Use the pH of a strong base to calculate the OH^- concentration, H^+ concentration and pH of the solution**
    Above
23. **Calculate the pH of a solution formed by mixing solutions of acids and / or bases**
    Calculate H^+ and OH^- concentration → calculate concentrations after reacting → Calculate pH

Stoichiometry

1. **State that the molar volume of all gases is 22.71 L at STP (0°C and 100 kPa), and use this to calculate the volumes or amount in moles of gases at STP**
   STP for calculating gas volumes is 0°C, when talking about anything else it is 25°C
2. **Use the mole concept and the molar volume to calculate quantities of gases as reactants of products in a chemical reaction at STP**
   $$n = \frac{V}{22.71}$$
   $$n = \frac{m}{M}$$
3. **Perform stoichiometric calculations in a range of contexts from across year 11, including limiting reagent calculations**

TOPIC 4 : CARBON CHEMISTRY & ENERGY

Structure & Naming

1. **Explain why carbon is able to form such a diverse range of compounds, using knowledge of bonding capacity**
   Carbon can make 4 covalent bonds as it has 4 valence electrons, allowing it to be the basis of a wide range of compounds
2. **Use molecular formulae, full structural formulae and condensed formulae to show the arrangement of atoms and bonds in covalent molecular substances, including hydrocarbons and haloalkanes**
3. **Define hydrocarbon, and categorise simple hydrocarbons into alkanes, alkenes and aromatic hydrocarbons (benzene)**
   Hydrocarbon: A compound consisting of only hydrogen and carbon atoms
   Alkane: Saturated - only C-C single bonds
   Alkenes: Has at least 1 C=C double bond
   Aromatics: Benzene (C_6H_6) (intermediate bonds between single and double) makes up part of the molecule
4. **Use the general formulae for alkanes and alkenes to recognise and categorise hydrocarbons**
   Alkanes: C_nH_{2n+2}
   Alkenes: C_nH_{2n}
5. **Recognise cycloalkanes and cycloalkenes, and understand that they have different general formulae from their non-cyclic counterparts**
   Cyclo- hydrocarbons join together in a ring-like shape
   Cycloalkanes: C_nH_{2n}
   Cycloalkenes: C_nH_{2n-2}
6. **Describe hydrocarbons as saturated or unsaturated, and explain the difference between these terms**
   Saturated: Only C-C single bonds, with the maximum possible number of atoms bonded to C atoms
   Unsaturated: Opposite of saturated
7. **Use IUPAC nomenclature to name straight and simple branched chain alkanes and alkenes (C_1 – C_8, one double bond per molecule)**
8. **Use IUPAC nomenclature to name straight chained haloalkanes (C_1 – C_8)**
   Use chloro-, bromo-, iodo- for halogen groups
9. **Draw and name structural isomers of alkanes and alkenes, specifically chain isomers and position isomers**
10. **Draw and name geometric isomers of alkenes, using cis and trans notation**
    Cis- = same side; Trans- = opposite side
11. **Describe the nature of bonding in alkanes, alkenes and benzene, using ideas of single and multiple bonds (π bonds), and delocalisation**
    Alkanes: Single π bonds
    Alkenes: One π bond, one σ bond
    Benzene: Intermediate bond length
12. **Explain why alkenes are able to form geometric isomers, whereas alkanes are not**
    Double bonds are non-rotational (atoms cannot rotate around them), while single bonds are rotational.
    For alkanes (only single bonds), the orientation of bonds can change easily, so bond orientation cannot be used to define isomers.
    Alkene double bonds cannot rotate, so molecules with different orientation around the double bond are isomers

Reactions of Hydrocarbons

1. **Explain the difference in chemical reactivity of alkanes, alkenes and benzene using discussion of bonding**
   Alkanes: Inert - only single sigma bonds which gives it a high activation energy
   Alkenes: Easily react with halogens and hydrogen halides as the alkene group has a pi bond which is very weak, and breaks easily
   Benzene: Inert - similar to alkanes. It doesn't have any pi bonds, instead its bonds

are an intermediate length between that of a single and double bond

2. **Write equations, and draw and name the products for substitution reactions of alkanes and benzene with halogens**
   Substitution: Halogen takes the place of a hydrogen atom
   Products will be a halogenoalkane/halogen-aromatic + Hydrogen halide
3. **Explain why substitution reactions of alkanes require visible / UV light to take place**
   High activation energy of alkanes due to them being saturated molecules means they need a source of external energy for the reaction to take place
4. **Recognise that substitution reactions of benzene require a catalyst**
   Substitution for benzene always requires a catalyst
5. **Identify, draw and name the products from single and multiple substitution reactions**
6. **Write equations, and draw and name the products of addition reactions of alkenes, with halogens, hydrogen halides and hydrogen**
   Addition: Two reactants form one product
   Alkenes always react in addition reactions; the pi bond breaks and two new atoms bond to the molecule
   Conditions:
      Halogens/Hydrogen Halides: None
      Hydrogenation (+H_2): Nickel catalyst, high temperature/pressure
      Hydration (Water): H_3PO_4 catalyst, high temperature/pressure
         Forms alcohol
7. **Describe how aqueous solutions of bromine and iodine can be used to test for unsaturation, and give observable outcomes**
   If there is an unsaturated compound, the solution will decolourise quickly as there are no conditions
   If it is a saturated hydrocarbon, it requires UV light so no reaction will be visible, or a minor change due to sunlight
   E.g. $Br_{2(aq)}$ + C_2H_4: "Orange solution added to colourless liquid, solution turns colourless leaving two immiscible colourless layers"
8. **Write equations, and draw and name the products of complete combustion reactions of hydrocarbons**
   Products: $H_2O_{(g)}$ (water vapour) + $CO_{2(g)}$ (Carbon dioxide)
9. **Given information about the products, write equations for the incomplete combustion reactions of hydrocarbons**
   Products: $H_2O_{(g)}$ (water vapour) + $CO_{(g)}$ (Carbon monoxide)
   Soot (C) may also be a product
   CO is a poisonous gas which is difficult to detect, named the "silent killer"
10. **Explain why incomplete combustion is undesirable when using hydrocarbons as fuels**
    Incomplete combustion:
    1. Produces CO which is poisonous and very dangerous
    2. Releases less energy as the enthalpy of C/CO is higher than that of CO_2
11. *Discuss the benefits and drawbacks of using hydrocarbons as fuels, including the effect on the environment. (SHE)*
12. ***Compare the effectiveness and suitability of different fuels, including hydrocarbons and biofuels (biogas, biodiesel, bioethanol), and suggest why certain fuels are used. (SHE)***
13. **Calculate percentage composition of a hydrocarbon from the relative atomic masses of the constituent species**
    If has things other than just H and C (although not a hydrocarbon), take away the mass of C/H to find the mass of the other element
14. **Use the mole concept and the law of conservation of mass to calculate quantities of reactants and products in a chemical reaction**
15. **Identify and calculate empirical formulae for hydrocarbons using percentage composition data and combustion analysis data, and relate this to the molecular formulae**
16. **Use molecular mass and empirical formulae calculations to establish molecular structure**

TOPIC 5 : ENERGY & RATE

Energy & Enthalpy

1. **State that kinetic theory is used to explain the macroscopic behaviour of gases using understanding of the molecular behaviour of particles**
   Kinetic theory tries to explain the behaviour of particles, and can be used to explain the properties of gases
2. **Describe the properties of an ideal gas, and describe the differences between an ideal gas and a real gas**
   Ideal Gas Properties:
   1. Particles move in constant random linear motion
   2. Volume of particles is negligible compared to volume of gas

3. Average E_K of particles is proportional to the temperature
4. No attraction/repulsion between particles
5. Collisions are elastic (no E_K lost when particles collide)

Differences:

- Real gas particles have volume
- Real gas particles have forces of attraction/repulsion (intermolecular forces)

3. Use kinetic theory to explain the behaviour of ideal gases, including diffusion and compressibility

Diffusion: Gases move in random motion, and have no attractive/repulsive forces between each other. Therefore if different gases are placed in the same container, they will readily diffuse through each other.

Compressibility: The volume of gas particles is negligible compared to the volume of the gas, therefore there is plenty of free space which they can be compressed into

Exert Pressure: As particles move in random linear motion, they will collide with their container's walls, exerting pressure

Pressure is a measure of the force exerted over a certain area

4. Describe the relationship between temperature, kinetic energy and velocity of particles in a qualitative fashion

As temperature increases, the average kinetic energy and velocity of particles will increase

5. Explain the origin of the Kelvin temperature scale, and state the meaning of the term absolute zero

Absolute Zero: The temperature where particles have no kinetic energy (no reactions will occur as there will be no collisions)

6. Convert temperatures between Celsius and Kelvin

°C = °K - 273.15

°K = °C + 273.15

7. Use kinetic theory to describe and explain the relationships between pressure, temperature and volume of an ideal gas in a qualitative fashion, (Boyle's Law, Charles' Law, and Pressure Law)

Boyle's Law: $PV = k$

Charles's Law: $V/T = k$

Pressure Law: Pressure exerted on the walls of a container by a gas is proportional to its temperature

8. Draw and interpret graphs that represent the relationship between pressure, temperature and volume

V/P both increase with T

9. State that the relationships between pressure, temperature, volume and number of moles can be combined to give the ideal gas equation, PV = nRT (Use of equation not required)

P: Pressure (kJ)

V: Volume (L)

n: Moles (mol)

R: Gas constant (8.314)

T: Temperature (°K)

10. Describe the internal energy of a system as the sum of the kinetic energy and the potential energy of the particles

Enthalpy (H): Chemical potential energy of a system at constant pressure

11. Describe the enthalpy of a system as a measure of the energy stored within a system, and state that it cannot be measured directly

Enthalpy cannot be measured directly - instead we measure the change in enthalpy using measurable energy changes such as heat/temperature

12. Describe the enthalpy change of a system as the heat energy exchange with the surroundings

Enthalpy Change: Heat energy exchanged with the surroundings

13. State that enthalpy and enthalpy change values are given at STP conditions

Enthalpy change is always measure at STP, and enthalpy is always given at STP

STP = 25°C/1atm = 298°K/100kPa

14. Describe and explain the enthalpy changes in a system in terms of stored (chemical potential) energy of reactants and products, and energy transfer in the form of heat, using the Law of Conservation of Energy

Law of Conservation of Energy: Energy cannot be created nor destroyed, but may be converted between different forms

Endothermic reactions: $E_K \rightarrow$ CPE

Exothermic reactions: CPE $\rightarrow E_K$

15. Describe observable changes in temperature, and use this evidence to establish whether a process is exothermic or endothermic

Endothermic: System/Surroundings get cooler

Exothermic: System/Surroundings get warmer

16. Explain the enthalpy changes in exothermic and endothermic processes in terms of energy input for breaking of existing bonds, and energy output from the

formation of new bonds

Endothermic: Energy required to break bonds > energy output from formation of bonds
Exothermic: Energy required to break bonds < energy output from formation of bonds

17. **Use thermochemical equations to represent the enthalpy changes in chemical reactions**

    Thermochemical Equation: Write ΔH to the right of the equation
    Endothermic: ΔH positive
    Exothermic: ΔH negative

18. **Draw and interpret enthalpy profile diagrams for exothermic and endothermic reactions using a quantitative scale for enthalpy, including accurate labels for reactants and products, activation energy, enthalpy change and transition state / activated complex**

    x-axis: Progress of reaction
    y-axis: Enthalpy (kJ)
    Activated Complex: Unstable arrangement of atoms with maximum CPE in which reactant bonds are breaking and product bonds are forming
    Activation energy: Reactants → Activated complex (vertical difference)
    Enthalpy Change: Reactants → Products (vertical difference)

19. ***Generate experimental temperature data from a reaction, to calculate the enthalpy change per mole of reaction (as stated in the chemical equation), given $\Delta H = mc\Delta T$, (c = specific heat capacity of solution, assumed to be 4.18 $JK^{-1}mol^{-1}$). (SIS)***

20. ***Evaluate outcomes from experiments involving enthalpy changes, identify sources of experimental error, and discuss the effect on enthalpy change values calculated. (SIS)***

21. ***Conduct an investigation into choice of fuels for a purpose; propose hypotheses, predict outcomes, collect reliable and valid data, from which conclusions can be drawn. (SIS)***

22. **Use enthalpy change data to compare the energy content or density of fuels, including biofuels**

    Divide ΔH by the moles of fuel to get the heat of combustion (ΔH_C). Comparing the heat of combustion tells you which fuel is the most efficient per mole
    Other comparison units may be kJ g^{-1} or MJ $tonne^{-1}$ for non-pure fuels

Rate of Reactions

1. **State that the rate of a chemical reaction can be quantified by measuring the rate of change in an observable quantity that indicates amount of reactant used up or product made**
    - $Rate\ of\ Reaction = \frac{Quantity\ of\ substance\ used/produced}{time}$
    - May measure:
        - Change in solution mass (if gas evolved)
        - Mass of precipitate formed
        - Colour change
        - pH
2. **Describe a range of methods to measure the rate of a reaction, including volume of gas produced, mass lost, colour change and changes to transparency**

    Volume of Gas: Upside-down cylinder filled with water, measure water displacement
    Mass Lost: Put beaker on a weighing scale and record the mass every x minutes
    Colour Change: Qualitative observations, record time taken for a particular colour to be achieved
    Transparency Change: Put beaker on a cross, record time taken until cross can no longer be seen

3. **Describe how to change the rate of a chemical reaction, using changes to reactant concentration, gas pressure, temperature, surface area of solid reactant, or use of a catalyst**
    - **↑Concentration/Pressure**: More particles in a given amount of space will result in more collisions → ↑Successful collisions + ↑Reaction rate
    - **↑Surface Area**: Reactions can only take place at the surface of solids (and any heterogeneous reaction). Increasing the surface area increases the particles available for collisions and reactions, therefore increasing collision rate and by extension reaction rate
    - **↑Temperature**: Increasing temperature increases the average kinetic energy of particles. This means 2 things: 1. Higher collision rate (more collisions), and 2. More collision energy (more successful collisions). Together, they greatly increase the reaction rate, with a 10°K/C rise in temperature doubling the rate of reaction. Higher collision energy is by far the main factor in increasing the reaction rate.
    - **Catalyst**: Chemical which increases the rate of reaction by providing an alternate

reaction pathway with a lower activation energy without any net consumption. Lower E_a means that a greater number of particles will have energy $\geq E_a$, and therefore more collsions will be successful

4. **State that a catalyst is a substance that speeds up a chemical reaction without being consumed**
5. ***Conduct experimental work safely, competently and methodically in order to collect valid and reliable data to measure the rate of reaction. (SIS)***
6. ***Represent rate of reaction data in tables and graphs, using correct units and symbols, in order to identify trends and any anomalous data. (SIS)***
7. ***Evaluate the effect of measurement error in numerical data. (SIS)***
   Measurement error may result in a higher/lower rate of reaction being calculated than the actual rate.
8. **Explain the factors that affect the rate of reaction using collision theory**
   Above
9. **Predict the outcome on the rate of reaction when conditions for a chemical reaction are altered using collision theory**
   Above
10. **State that the activation energy for a reaction is the minimum amount of energy required for particles to react on collision**
    Activation Energy (E_a): Minimum amount of energy particles must collide with to break reactant bonds for a reaction to occur
11. **Describe how the size of the activation energy affects the rate of reaction**
    Lower activation energy means more collisions will have collision energy $\geq E_a$, and therefore more collisions will be successful, increasing the rate of reaction. Opposite occurs for reactions with a higher activation energy
12. **Explain how the number and strength of bonds to be broken affects the size of the activation energy**
    More bonds and stronger bonds = higher activation energy
    Less bonds and weaker bonds = lower activation energy
13. **Describe and explain the effect of using a catalyst on the rate of reaction using collision theory**
    A catalyst provides a reaction pathway with a lower activation energy, therefore more collisions will have energy $\geq E_a$ and more successful collisions will occur, increasing the reaction rate
14. **Draw and interpret energy profile diagrams for catalysed and uncatalysed reactions.**
    Use a dotted line + label catalysed pathway
15. ***Justify the use of catalysts in industry, to increase the rate of a reaction that would otherwise be uneconomically slow. (SHE)***
    In industry, reactions must be sped up to make them viable to produce products. Increasing temperature/pressure is very expensive and also harmful to the environment, therefore catalysts are used instead.

    E.g. The Haber Process ($N_2 + 3H_2 \rightarrow 2NH_3$)
    Iron is used as a catalyst in the Haber Process, which is used to produce ammonia for fertilizers and explosives. Uncatalysed, the reaction requires temperatures of approximately 3000°C, but with the catalyst, it can take place at just 500°C, which is much cheaper.
16. ***Describe how catalysts can be used to reduce negative impact on the environment, for example in catalytic converters. (SHE)***
    Catalytic Converters in Cars
    Platinum, Palladium and Rhodium are used in catalytic converters which convert toxic gases (CO, NO_X, unburnt hydrocarbons) into harmful gases (CO_2, N_2, H_2O). They are in a very finely powdered form in a honeycomb monolith to increase their exposed surface area to increase their efficiency, which is especially important as they are expensive metals, and the gases they need to convert are in the catalytic converter for less than half a second.

Experiments

Tuesday, 11 August 2020 9:15 PM

Pearson

Types of Data

- **Qualitative Data**: Can be observed and sorted into groups but not measured
  - E.g. Brightness, colour, type of material
- **Quantitative Data**: Can be measured
  - E.g. Mass, volume, pH, temperature, time

Experiment Labels

- **Validity**: Whether an experiment is testing the set hypothesis and aims. Only one variable may change in a valid experiment
- **Reliability**: If an experiment is repeated, the results should be consistent
- **Accuracy**: Average of results is correct (minimise systematic error)
- **Precision**: All results are close together (minimise random error)

Uncertainty and Error

- **Uncertainty**: How far the experimental value may be from the true value
  - E.g. ±1%
- **Systematic Errors**: Always above, or always below, the true value
  - E.g. Not subtracting the mass of a beaker when weighing a substance
- **Random Error**: May be higher or lower than the true value
  - Will occur whenever using a measuring device (e.g. Measuring cylinder)
- **Mistakes**: Dumb things such as recording the wrong number or spilling a solution
  - Not classified as errors
  - Do not include in calculations
- Improve reliability by repeating trials

Atomic Structure

Wednesday, 3 June 2020 9:01 PM

Atomic Model

History

Name	Date	Model	Experiment	Theory	Discovery
John Dalton	1803			Atomic Theory 1. Elements are composed of extremely small particles called atoms which are indivisible 2. All atoms of the same element are identical; same size, mass and chemical properties 3. Atoms cannot be created nor destroyed nor changed into a different element during a reaction 4. A chemical reaction involves only separation, combination or rearrangement of atoms 5. Compounds are formed when atoms of different elements combine in a whole fixed number ratio	Ideal Gas Law
J.J. Thompson	1897 1904	Plum-Pudding Positively charged sphere with electrons scattered throughout	Cathode Ray	Neutrons - unaccounted mass in the nucleus	Electrons: small, negative
Ernest Rutherford	1911	Nuclear Atom mostly empty space, electrons orbit nucleus	Gold Foil		Nucleus: small, dense, positive
Niels Bohr	1913	Bohr Electrons orbit in fixed orbital radii (shells)		Electrons exist in discrete energy levels with no intermediate energies	Energy levels of single-electron-atoms
Erwin Schrodinger	1926	Quantum Mechanical Model Electron cloud Electrons have wave and particle nature			Electrons don't orbit in discrete radii
James Chadwick	1932			Neutrons same mass as protons, no charge	Neutrons

Subatomic Particles

Particle	Relative Mass	Charge	Location
Proton	1	+1	Nucleus
Neutron	1	0	Nucleus
Electron	$\frac{1}{1836}$	-1	Shells

Definitions

- **Atom:** The smallest unit of matter that can exist on its own; it is chemically indivisible
- **Element:** The simplest kind of substance that can exist; it cannot be broken down into anything simpler
- **Compound:** Two or more different elements chemically combined in a specific whole number ratio
- **Molecule:** A particle consisting of two or more atoms covalently bonded
- **Pure Substance:** Matter with fixed/definite composition that cannot be separated by physical means; Well defined physical and chemical properties
- **Mixture:** Two or more pure substances physically combined in no fixed ratio
  - Homogeneous Mixture: A mixture with uniform composition throughout
  - Heterogeneous Mixture: A mixture with non-uniform composition; Constituents are physically separated
- **Ion:** An atom/group of atoms with positive/negative charge due to loss/gain of electrons
- **Isotope:** Atoms of the same element with the same number of protons but a different number of neutrons
  - Same chemical properties as they have the same electron configuration
- **Relative Atomic Mass:** Mean mass of all atoms of an element relative to 1/12 the mass of a Carbon-12 atom
- **Mole:** A precisely defined quantity of matter equal to Avogadro's number ($6.022 * 10^{23}$) of particles

Electron Configuration

Subshells

Shell	Subshells	Orbitals	Max Electrons
n = 1	$s_{(2\ electrons)}$	1	2
n = 2	$s_{(2\ electrons)}$ $p_{(6\ electrons)}$	1 3	8
n = 3	$s_{(2\ electrons)}$ $p_{(6\ electrons)}$ $d_{(10\ electrons)}$	1 3 5	18

- Max number of electrons = $2n^2$
- Electrons fill subshells from the lowest energy to the highest energy
- E.g. $_{20}Ca$: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$

Orbitals

- Orbital: A region around the nucleus that can hold up to 2 electrons with opposite spins
  - Opposite spin prevents repulsion
- Different orbitals (s/p/d/f) differ in energy and shape, occupying different regions of space
- s-orbital: spherical (draw as circle)
- p-orbital: hour=glass (draw as figure-8)

Absorption/Emission Spectra

- **Absorption Spectra:** The spectrum of electromagnetic radiation produced when a continuous spectrum is transmitted through an element
  - Black lines where the element absorbed the specific frequencies
- **Emission Spectra:** The spectrum of electromagnetic radiation produced by an element when its electrons return to their ground state after being excited
  - Fully black spectrum with specific frequencies visible
- Every element has a unique absorption/emission spectra

How it Works

- Electrons can absorb energy (heat, electric discharge or electromagnetic radiation) and 'jump' to a higher energy state (Bohr's model states they can only exist in discrete orbits)
- As they are unstable there, they emit energy (as electromagnetic radiation) and return to their ground state

Common Ion Colours

H^+	Violet
K^+	Purple
Na^+	Yellow
Li^+	Red
Ca^{2+}	Orange/Red
Sr^{2+}	Orange/Red
Cu^{2+}	Blue/Green
Ba^{2+}	Green

Atomic Absorption Spectroscopy

1. Sample vapourised
2. Light of specific frequencies are shone through it, light that passes through is recorded by a detector
3. Missing frequencies can identify the element (absorption spectra), and intensity of absorption can determine the concentration

Flame Tests

- Excite electrons using heat, they jump up energy states
- Photons of specific frequencies emitted when they return to their ground state
- Combination of frequencies produces a visible colour, can be viewed using spectroscope

  Limitations
  - Only a small number of metals may be identified as heat isn't enough to excite many element's electrons (higher ionisation energy)
  - Hard to distinguish between similar colours
  - Contamination affects colour produced

Periodic Trends

Ionisation Energy

- **First Ionisation energy:** The amount of energy required to remove one mole of electrons from one mole of an atom in its natural gaseous state
  - $X_{(g)} \rightarrow X^+_{(g)} + e^-$
- Increases up a group (less shells = closer to nucleus + less shielding) and across a period (higher nuclear charge = more attraction to valence electrons)
  - Helium highest ionisation energy

Electronegativity

- **Electronegativity:** The tendency of an atom in a molecule to attract bonding electrons towards itself
- Same trend as ionisation energy, same reasons
  - Fluorine highest ionisation energy - Helium not counted as noble gases generally don't bond

Atomic Radius

- Increases going down a group (more shells = more shielding + distance)
- Decreases across a period (higher nuclear charge = more attraction)

Reactivity

- Non-metals (16&17): Top-Right (F) most reactive due to higher electronegativity
- Metals (1&2): Bottom-Left (Fr) most reactive due to lower ionisation energy

John Dalton

Atomic Theory (1803)

Atomic Theory

1. Elements are composed of extremely small particles called atoms which are indivisible
2. All atoms of the same element are identical; same size, mass and chemical properties
3. Atoms cannot be created nor destroyed nor changed into a different element during a reaction
4. A chemical reaction involves only separation, combination or rearrangement of atoms
5. Compounds are formed when atoms of different elements combine in a whole fixed number ratio

Problems with it

i. Atoms *are* divisible (protons, neutrons, electrons)
ii. Atoms of the same element can have different masses (isotopes)

J.J. Thomson

Electron (1897)
Plum Pudding Model (1904)

Summary

- Discovered **electrons** in 1897 using the cathode ray experiment
- **Cathode ray:** A stream of electrons produced from cathode (negative terminal) of vacuum tube
- Proposed the **plum pudding model** in 1904
- Estimated mass of one electron to be < 1000th of a Hydrogen atom
- Concluded that atoms have tiny, negatively charged particles inside them

Plum Pudding Model

- Numerous tiny low-mass negatively-charged electrons embedded within a large positively-charged sphere

Ernest Rutherford

Nucleus & Nuclear Model (1911)

Gold Foil Experiment

- Fired a beam of alpha particles (He^{2+}) at a very thin sheet of gold
- Some of them were deflected completely, if the Plum Pudding Model was right they would all have gone straight through

Nuclear Model

- There had to be a tiny, dense and heavy source of positive charge to account for the particles being deflected to the extent that they were
- Proposed the atom consisted of mostly empty space occupied by only the low-mass negatively charged electrons
- Electrons orbited tiny central region called the nucleus
- Nucleus made up nearly all the mass, and all the positive charge of the atom

Niels Bohr

Bohr Model (1913)

Problems with Rutherford's Model

- Could not explain emission/absorption spectra
- Did not explain idea of electrons orbiting the nucleus, as they would lose energy and spiral into the nucleus

The Bohr Model

- Theorised that the electronic structure was quantised
  - I.e. Electrons could only have specific energies and no intermediate energies
  - Orbits of larger radii corresponded to higher energy levels (shells)
  - Movement between energy levels could occur through absorption/emission of energy in form of electromagnetic radiation
- Explained emission spectra of hydrogen
  - Only worked for single electron particles (H, He^+, Li^{2+}, etc.)

Erwin Schrödinger

Quantum Mechanical Model (1926)

Quantum Mechanical Model

- Used mathematical equations to determine the likelihood of finding an electron in a particular position
- Doesn't define electron path, instead predicts odds of the location of the electron
- Portrayed as a nucleus surrounded by an electron cloud; more dense = more chance of finding an electron there
- Currently accepted model

James Chadwick

Neutron (1932)

Experiment

- Identified neutrons by bombarding beryllium with alpha particles

Atomic Model

Wednesday, 3 June 2020 9:06 PM

Development/History

Name	Date	Model	Experiment	Theory	Discovery
John Dalton	1803			Atomic Theory 1. Elements are composed of extremely small particles called atoms which are indivisible 2. All atoms of the same element are identical; same size, mass and chemical properties 3. Atoms cannot be created nor destroyed nor changed into a different element during a reaction 4. A chemical reaction involves only separation, combination or rearrangement of atoms 5. Compounds are formed when atoms of different elements combine in a whole fixed number ratio	Ideal Gas Law
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Subatomic Particles

Particle	Relative Mass	Charge	Location
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Electron	$\frac{1}{1840}$	-1	Shells

Definitions

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- **Compound:** Two or more different elements chemically combined in a specific whole number ratio
- **Atom:** The smallest unit of matter that can exist on its own; it is chemically indivisible
- **Isotope:** Atoms of the same element with the same number of protons but a different number of neutrons
    - Same chemical properties as they have the same electron configuration
    - Different physical properties
- **Molecule:** A particle consisting of two or more atoms covalently bonded
- **Ion:** An atom/group of atoms with positive/negative charge due to loss/gain of electrons
- **Relative Atomic Mass:** Mean mass of all atoms of an element relative to 1/12 the mass of a Carbon-12 atom
    - No units as it is relative
- **Mole:** A precisely defined quantity of matter equal to Avogadro's number ($6.022 * 10^{23}$) of particles
    - $n = \frac{\text{number of particles}}{\text{Avogadro's number}}$
    - $n = \frac{m}{M}$

Full Timeline

Wednesday, 3 June 2020 9:37 PM

John Dalton

Atomic Theory (1803)

Atomic Theory

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2. All atoms of the same element are identical; same size, mass and chemical properties
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4. A chemical reaction involves only separation, combination or rearrangement of atoms
5. Compounds are formed when atoms of different elements combine in a whole fixed number ratio

General Info

- Work based on law of conservation of mass and law of constant composition
- Parts of the theory were proved incorrect later due to discovery of subatomic particles
    i. Atoms *are* divisible (protons, neutrons, electrons)
    ii. Atoms of the same element can have different masses (isotopes)

J.J. Thomson

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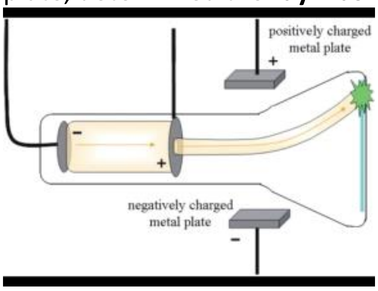
Summary

- Discovered **electrons** in 1897 using the cathode ray experiment
- **Cathode ray:** Rays produced from cathode (negative terminal) of vacuum tube
- Proposed the **plum pudding model** in 1904

Cathode Ray Experiment

- When power source connected to two pieces of metal, ray shot from negatively charged metal to positively charged metal, some of it went through the hole in the positively charged metal plate
- To check if the stuff the ray was made out of was charged, he put charged plates on either side of the tube
- It curved away from the negatively charged plate and towards the positively charged plate, determined the **ray was made of negatively charged particles**



Conclusions/Discoveries

- Deduced charge to mass ratio of cathode ray particles (electrons)
- Estimated mass of one cathode ray particle (electron) to be < 1000th of a Hydrogen atom
- Swapped metals out for different elements, got same result - showed that the element didn't affect the particles produced)
- Concluded that atoms have tiny, negatively charged particles inside them (disproved part 1 of atomic theory)

Plum Pudding Model

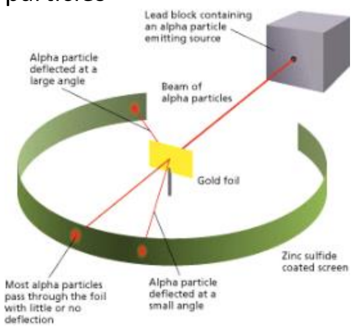
- Numerous tiny negatively-charged electrons embedded within a large positively-charged sphere
- Large sphere accounted for nearly all of its mass

Ernest Rutherford

Nucleus & Nuclear Model (1911)

Gold Foil Experiment

- Fired a beam of alpha particles (He^{2+}) at a very thin sheet of gold
- According to the plum pudding model the particles should pass straight through with little/no deflection as the positive charge of the atom was uniformly distributed and not dense
- Most particles passed through, but some ($\approx \frac{1}{20000}$) were deflected significantly
- He was surprised - wrote it was "as if you fired a 15 inch shell into a piece of tissue paper and it came back and hit you"
- Plum Pudding model could not be right as it couldn't account for heavily deflected alpha particles



Conclusions/Discoveries

- There had to be a tiny, dense and heavy source of positive charge to account for the particles being deflected to the extent that they were
- Discovered the nucleus

Nuclear Model

- Proposed the atom consisted of mostly empty space occupied by only the low-mass negatively charged electrons
- Electrons orbited tiny central region called the nucleus
- Nucleus made up nearly all the mass, and all the positive charge of the atom

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Neutron (1932)

Experiment

- Identified neutrons by bombarding beryllium with alpha particles

Electron Configuration

Friday, 10 July 2020 11:47 AM

Subshells

Shell	Subshells	Orbitals	Max Electrons
n = 1	s(2 electrons)	1	2
n = 2	s(2 electrons) p(6 electrons)	1 3	8
n = 3	s(2 electrons) p(6 electrons) d(10 electrons)	1 3 5	18

- Max number of electrons = $2n^2$
- Electrons fill subshells from the lowest energy to the highest energy
- E.g. $_{20}Ca$: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$

Orbitals

- Orbital: A region around the nucleus that can hold up to 2 electrons with opposite spins
    - Opposite spin prevents repulsion
- Different orbitals (s/p/d/f) differ in energy and shape, occupying different regions of space
- s-orbital: spherical (draw as circle)
- p-orbital: hour=glass (draw as figure-8)

Absorption/Emission Spectra

- **Absorption Spectra:** The spectrum of electromagnetic radiation produced when a continuous spectrum is transmitted through an element
    - Black lines where the element absorbed the specific frequencies
- **Emission Spectra:** The spectrum of electromagnetic radiation produced by an element when its electrons return to their ground state after being excited
    - Fully black spectrum with specific frequencies visible
- Every element has a unique absorption/emission spectra

How it Works

- Electrons can absorb energy (heat, electric discharge or electromagnetic radiation) and 'jump' to a higher energy state (Bohr's model states they can only exist in discrete orbits)
- As they are unstable there, they emit energy (as electromagnetic radiation) and return to their ground state

Common Ion Colours

H^+	Violet
Li^+	Red
Na^+	Yellow
K^+	Purple
Ca^{2+}	Orange/Red
Sr^{2+}	Orange/Red
Cu^{2+}	Blue/Green
Ba^{2+}	Green

Atomic Absorption Spectroscopy

1. Sample vapourised

2. Light of specific frequencies are shone through it, light that passes through is recorded by a detector
3. Missing frequencies can identify the element (absorption spectra), and intensity of absorption can determine the concentration

Flame Tests

- Excite electrons using heat, they jump up energy states
- Photons of specific frequencies emitted when they return to their ground state
- Combination of frequencies produces a visible colour, can be viewed using spectroscope

Limitations

- Only a small number of metals may be identified as heat isn't enough to excite many element's electrons (higher ionisation energy)
- Hard to distinguish between similar colours
- Contamination affects colour produced

Periodic Trends

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Reactivity

- Non-metals (16&17): Top-Right (F) most reactive due to higher electronegativity
- Metals (1&2): Bottom-Left (Fr) most reactive due to lower ionisation energy

Ionisation Energy

- **Ionisation energy:** The amount of energy required to remove one mole of electrons from one mole of an atom in its natural gaseous state
  - $X_{(g)} \rightarrow X^{+}_{(g)} + e^{-}$
- Increases up a group (less shells = closer to nucleus + less shielding) and across a period (higher nuclear charge = more attraction to valence electrons)
  - Helium highest ionisation energy

Electronegativity

- **Electronegativity:** The tendency of an atom in a molecule to attract bonding electrons towards itself
- Same trend as ionisation energy, same reasons
  - Fluorine highest ionisation energy - Helium not counted as noble gases generally don't bond

Nanomaterials

Wednesday, 3 June 2020 9:01 PM

Introduction

- Nanomaterials are materials with at least one dimension in the nanoscale
- Nanoscale = 1-100nm
    - 1nm = 10^{-9}m (billionth of a metre)
- Nanomaterials have different properties to their bulk material due to increased surface area to volume ratio, so surface (quantum) effects are more prominent
    - Differ in: optical, thermal, strength, density, stiffness
- **Nanocomposites** are combination of bulk materials and nanomaterials, often to enhance a property

Carbon Nanomaterials

- **Allotrope:** Different physical forms in which an element can exist
- **Fullerenes:** Carbon allotropes where each atom has 3 covalent bonds and one delocalised electron forming a sea of delocalised electrons
- **Graphene:** A 2D layer of carbon atoms
    - Very strong, light, good conductor
    - Replaces silicon in lots of technologies, desalination plants, used in composite materials
- **Carbon Nanotubes (CNTs)** are 'rolled up' graphene, diameter in nanoscale but unlimited length, single or multi-walled
    - 300 x strength of steel
    - 1/4 x density of steel
    - Semiconductor to superconductor based on dimensions
    - **Uses:** CPUs, bullet-proof vests, cars, planes, sport equipment

Other Uses/Examples

- **Dendrimers:** Nanodevices which can carry several molecules to find, target and kill cancer cells; leaves healthy tissue untouched
- Carbon black + rubber = car tyres; stops build-up of static electricity, increases strength & durability
- TiO_2 & ZnO_2 in sunscreen: Invisible vs white sunscreen
- Silver: Used in medical equipment as it kills bacteria
- Iron: Cure groundwater of CCl_4
- Au & Ag: Stained-glass windows (colour changes)

Creation

- **Top-Down:** Start with larger material then cut away selectively or grind it down to desired shape/size
    - E.g. Mechanical Grinding
    - Large quantities, relatively cheap, fairly uniform result
    - Sunscreen, CPU chips
- **Bottom-Up:** Build up atom by atom/molecule by molecule
    - E.g. Dip-Pen Lithography
    - Can create complex structures
    - Not economically viable for commercial use

Safety

- Can penetrate cell membrane - human immune system can break it down, but don't know whether other organisms can
- Enters environment when we wash clothes and hands, go to beach, etc. - enters air and skin through cell membrane - unwanted/dangerous reactions?
- Potential danger by breathing in/applying cosmetics

Separation Techniques

Monday, 17 February 2020 2:11 PM

Sieving

- Heterogeneous mixtures with different sized constituents
- Sieves have meshed/perforated bottoms which only allow particles of a specific size to pass through it
- E.g. Small particles from larger particles

Filtration

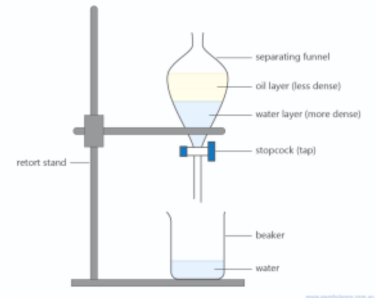
- Mixtures containing a solid suspended in a liquid
- Filter paper has small pores which allow the liquid to pass through, but aren't large enough for solid particles to pass through
- Filtered liquid is called the filtrate, solid is called the residue
- E.g. $NaCl/CuCl_2$, Charcoal/NaCl

Decanting

- Heterogeneous mixtures containing a solid suspended in a liquid OR two immiscible liquids
- Wait and allow the constituents to settle and separate by density
- A centrifuge can speed up the separation by simulating an increase in the force of gravity
- Lighter fluid is poured off, leaving the heavier fluid or solid behind (usually a small amount of the lighter fluid will be left behind)
- E.g. Oil/Water, Dirt/Water, Wine (sediment from fermentation/wine), Cream/Milk

Separation Funnel (Diagram)

- Heterogeneous mixture of two immiscible liquids
- Mixture is put into the separating funnel, allowed to stand until they separate (according to density)
- Opening of the stop-cock (tap), allows the denser layer comes out to be collected in a beaker
- When the denser liquid has completely run off, the stop-cock is closed
- Lighter liquid collected in another beaker by opening the stop-cock again
- E.g. Oil/Water



Separating Funnel

Distillation (Diagram)

- Homogeneous mixtures
- Based on a difference in boiling points
- When heated, lower BP substance evaporates first, resulting in a vapour which can be condensed later
- Some of the higher BP substance will also boil, not a pure substance
- Result called distillate

Fractional Distillation (Basic Diagram)

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Crystallisation

- Results in the formation of solid particles containing the dissolved solution
- As one substance evaporates, the dissolved substance comes out of the solution and collects as crystals
- Produces highly pure solids (e.g. Rock Candy)

Mass Spectrometry